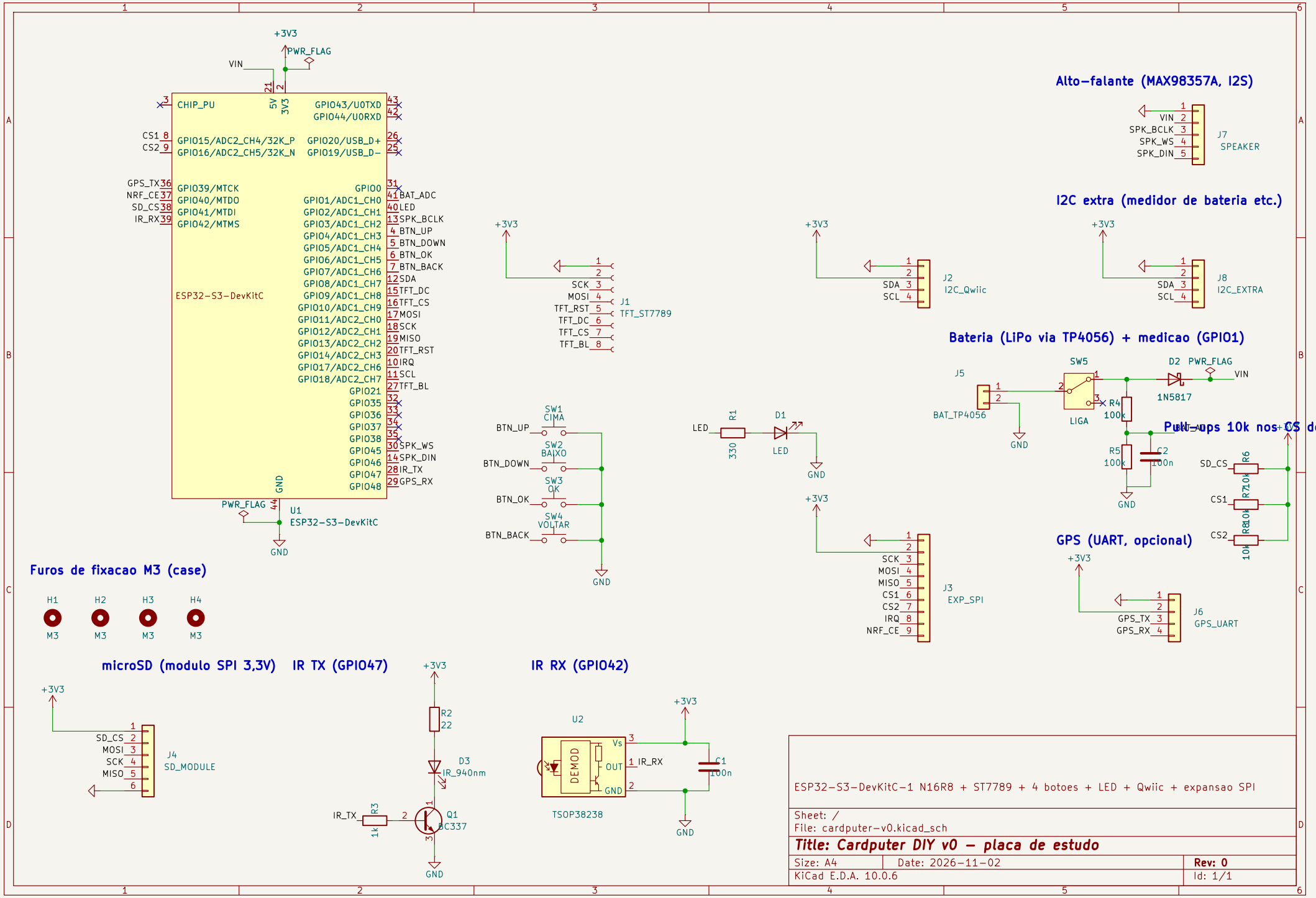

+3V3
PWR_FLAG
VIN
CHIP_PU
5V
3V3
GPIO43/U0TXD
GPIO44/U0RXD
CS1
CS2
GPIO15/ADC2_CH4/32K_P
GPIO16/ADC2_CH5/32K_N
GPIO20/USB_D+
GPIO19/USB_D-
GPS_TX
NRF_CE
SD_CS
IR_RX
GPIO39/MTCK
GPIO40/MTDO
GPIO41/MTDI
GPIO42/MTMS
GPIO0
GPIO1/ADC1_CH0
GPIO2/ADC1_CH1
GPIO3/ADC1_CH2
GPIO4/ADC1_CH3
GPIO5/ADC1_CH4
GPIO6/ADC1_CH5
GPIO7/ADC1_CH6
GPIO8/ADC1_CH7
GPIO9/ADC1_CH8
GPIO10/ADC1_CH9
GPIO11/ADC2_CH0
GPIO12/ADC2_CH1
GPIO13/ADC2_CH2
GPIO14/ADC2_CH3
GPIO17/ADC2_CH6
GPIO18/ADC2_CH7
GPIO21
GPIO35
GPIO36
GPIO37
GPIO38
GPIO45
GPIO46
GPIO47
GPIO48
BAT_ADC
LED
SPK_BCLK
BTN_UP
BTN_DOWN
BTN_OK
BTN_BACK
SDA
TFT_DC
TFT_CS
MOSI
SCK
MISO
TFT_RST
IRQ
SCL
TFT_BL
SPK_WS
SPK_DIN
IR_TX
GPS_RX
ESP32-S3-DevKitC
GND
PWR_FLAG
U1
ESP32-S3-DevKitC
Alto-falante (MAX98357A, I2S)
VIN
SPK_BCLK
SPK_WS
SPK_DIN
J7
SPEAKER
I2C extra (medidor de bateria etc.)
SDA
SCL
J8
I2C_EXTRA
+3V3
SCK
MOSI
TFT_RST
TFT_DC
TFT_CS
TFT_BL
J1
TFT_ST7789
J2
I2C_Qwiic
Bateria (LiPo via TP4056) + medicao (GPIO1)
J5
BAT_TP4056
SW5
LIGA
D2
PWR_FLAG
1N5817
R4
100k
R5
100k
C2
100n
BAT_ADC
Pull-ups 10k nos CS do SPI
SD_CS
CS1
CS2
R6
R7 10k
R8 10k
10k
SW1
CIMA
BTN_UP
SW2
BAIXO
BTN_DOWN
SW3
OK
BTN_OK
SW4
VOLTAR
BTN_BACK
LED
R1
330
D1
LED
SCK
MOSI
MISO
CS1
CS2
IRQ
NRF_CE
J3
EXP_SPI
GPS (UART, opcional)
GPS_TX
GPS_RX
J6
GPS_UART
Furos de fixacao M3 (case)
H1
H2
H3
H4
M3
microSD (modulo SPI 3,3V)
SD_CS
MOSI
SCK
MISO
J4
SD_MODULE
IR TX (GPIO47)
R2
22
D3
IR_940nm
IR_TX
R3
1k
Q1
BC337
IR RX (GPIO42)
U2
DEMOD
Vs
OUT
GND
IR_RX
C1
100n
TSOP38238
ESP32-S3-DevKitC-1 N16R8 + ST7789 + 4 botoes + LED + Qwiic + expansao SPI
Sheet: /
File: cardputer-v0.kicad_sch
Title: Cardputer DIY v0 — placa de estudo
Size: A4
Date: 2026-11-02
Rev: 0
KiCad E.D.A. 10.0.6
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